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NO EAR LEFT BEHIND:

WIRELESS EARBUDS FOR LOW-COST HEARING SCREENING

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Hearing loss is particularly harmful for language acquisition and neuro-development if it is left undetected in early childhood. Newborn hearing screening technologies using otoacoustic emissions (OAE) rely on detecting soft sounds generated by a healthy cochlea. High-income countries like the United States frequently implement hearing screening programs for every child at birth. However, such universal hearing screening is significantly less common in low- and middle-income countries, partly due to the conventional wisdom that the test requires sensitive and expensive acoustic hardware that costs thousands of dollars.

In our MobiSys paper[1], we present the first wireless earbud design for low-cost hearing screening. Our hardware-software system reliably performs hearing screening using low-cost acoustic hardware, while being in the form-factor of a wireless earbud (Figure. 1a,b). The earbud hardware is designed to work across a large age range from newborns to adults. By developing low-cost and open-source wearable technology, our work may help address global health inequities in hearing screening by democratizing these medical devices.

You might wonder how a hearing screening is performed in a young infant who is not able to respond to sounds in the same way as an adult. The goal of the test is to check the health of the cochlea, which is the coiled shape organ in the inner ear that is often affected by hearing loss (Figure 1c). The

cochlea is lined with hair cells (Figure 1d) that are highly sensitive to sounds and vibrate in response to anything that we hear. The fascinating thing is that while we think of the ear as a microphone that receives sound, it also acts as a speaker that generates sound. In particular, the way this is done

is by the vibration of the cochlea hair cells, which creates extremely soft sounds that are often inaudible to the human ear and require sensitive sensors to pick them up. In fact, it is precisely because such test equipment needs to be so sensitive that it often costs thousands of dollars and is a

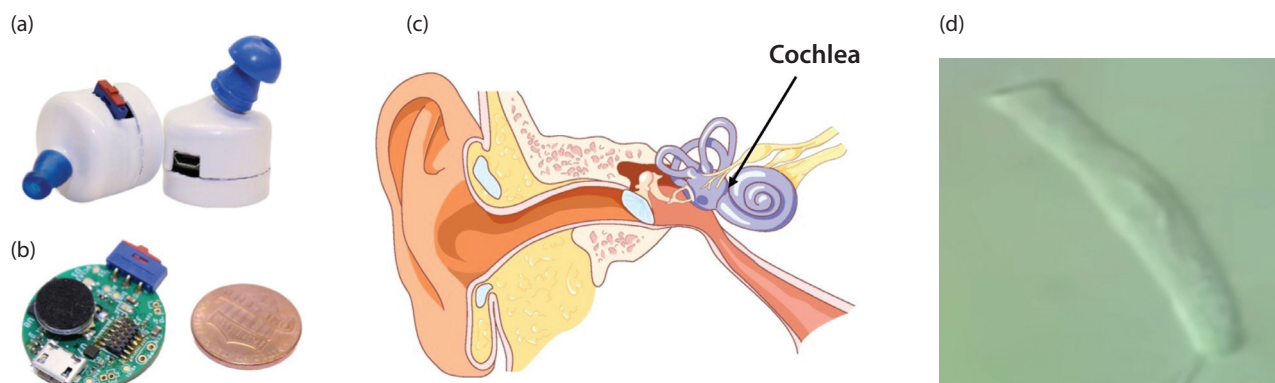


FIGURE 1. OAEbuds is a low-cost wireless earbud that can perform hearing screening by detecting otoacoustic emissions (OAE) from the cochlea. (a) The 3D-printed enclosure with pediatric and adult earbud tips. (b) OAEbud circuit board beside a penny for size comparison. (c) Location of the cochlea in the inner ear, which is critical for hearing. (d) Cochlea hair cells vibrate in response to sounds. These minute motions generate very soft sounds like a speaker (Source: <https://www.youtube.com/watch?v=Xo9bwQuYrRo>)

reason that limits hearing screening in low- and middle-income countries. Furthermore, in these countries, getting access to test equipment often requires travel to an urban setting and long wait times, which significantly limits access to hearing care. In an effort to increase access to hearing screening, we make the following contributions:

- We introduce *OAEbuds*, first wireless earbuds hardware design that can perform low-cost hearing screening by detecting signals from the cochlea.
- We create a wireless sensing protocol that combines frequency modulated continuous wave signals and wideband pulses to separate unwanted reflections and echoes from the ear canal from the signals from the cochlea.
- We performed a clinical study across two sites on pediatric and adult patients, and our system achieved accuracies similar to a \$5000 FDA-cleared medical device.

BY DEVELOPING LOW-COST AND OPEN-SOURCE WEARABLE TECHNOLOGY, OUR WORK MAY HELP ADDRESS GLOBAL HEALTH INEQUITIES IN HEARING SCREENING

SYSTEM DESIGN

Building this system requires addressing three key technical challenges. Firstly, low-cost microphones are not optimized to detect the soft sounds from the cochlea and are not as sensitive as the expensive microphones used in traditional medical devices. Secondly, the ear canal, eardrum, and wireless earbud case create unwanted acoustic reflections that will overlap with the sounds from the cochlea (Figures 2a,b). It is important to accurately distinguish the soft sounds generated by the cochlea from these unwanted reflections, as the reflections are stronger and can overwhelm it. Thirdly, the system needs to be robust when tested across different ear canal shapes that differ between patient age.

We address these challenges by taking advantage of two important properties of the cochlear emissions. The first property is related to how the unwanted reflections from the ear canal will arrive at the microphone first, and only after that will the sounds generated by the cochlea be recorded (Figure 2b). The second property is related to the curled geometry of the cochlea, which affects how the emissions are generated. Specifically, the beginning of the coil is responsive to high frequency sounds, while the end of the coil is responsive to low frequency sounds. When sound enters the cochlea, a high frequency response is first emitted, followed by lower frequency responses. This response pattern is entirely different from a reflection.

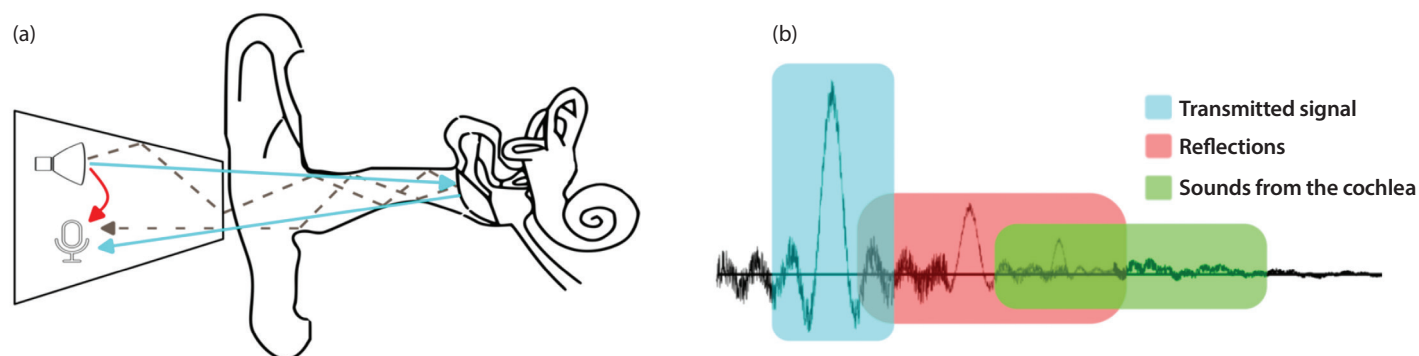


FIGURE 2. System design challenges: (a,b) The ear canal, eardrum, and earbud case create significant unwanted reflections that are louder than the sounds generated by the cochlea.

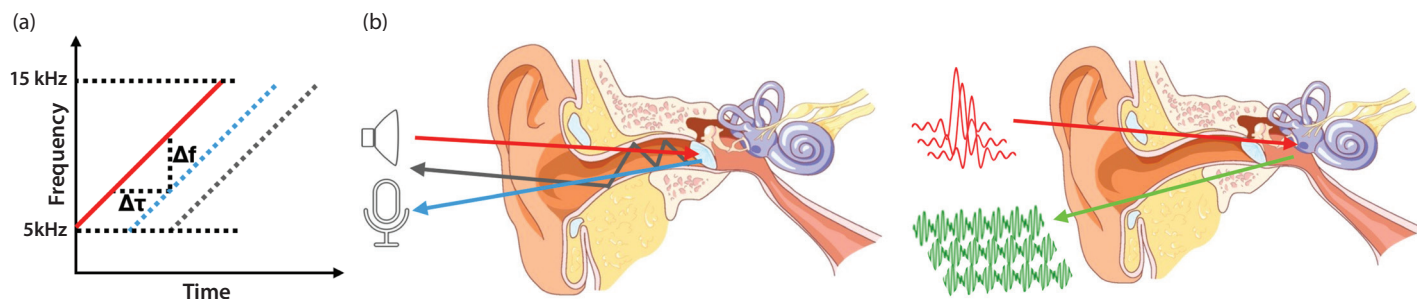


FIGURE 3. OAEbuds wireless sensing approach: (a) Linear frequency modulated chirps are sent into the ear to estimate delays caused by reflections from the ear canal. (b) A train of wideband pulses are sent into the ear to cover the range of frequencies important for hearing screening, and the cochlea's responses to these pulses are combined.

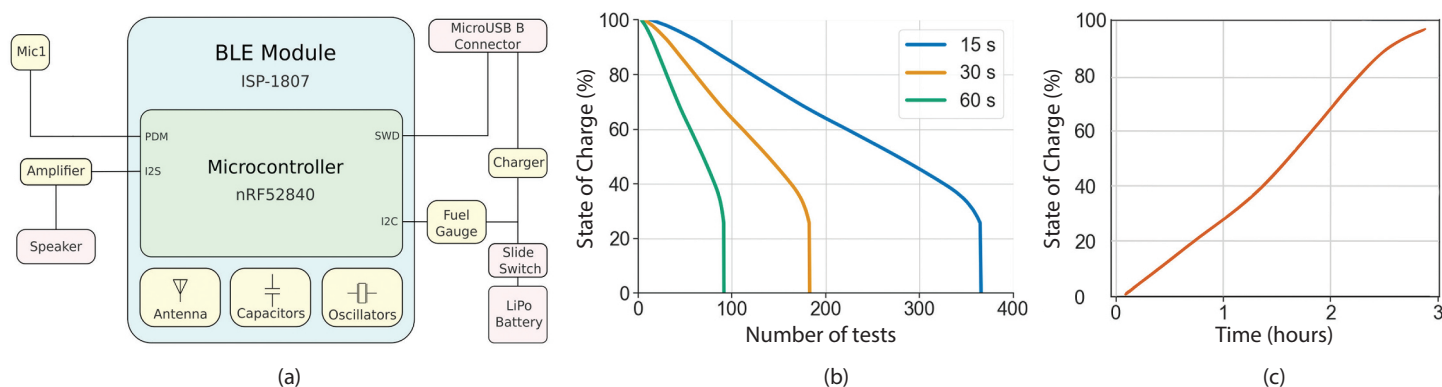


FIGURE 4. OAEbuds hardware design: (a) OAEbuds hardware design. (b) Number of tests that can be performed on a single charge for tests of different durations. (c) Time required to charge an OAEbud via micro-USB.

We develop a two-step protocol based on these properties. The first step as shown in Figure 3a is to estimate the duration over which the reflections last for. By estimating this delay, we can ignore the portion where the reflections are strong and focus on the time when the cochlea can be more easily heard. To do this, we send frequency modulated signals, which are signals that increase in frequency linearly over time. In our case, the signal increases from 5 to 15 kHz. After sending these signals into the ear, reflections come back at different time delays. As it turns out, each time delay is associated with a frequency shift that can be calculated after performing a Fast Fourier Transform.

In the second step of the protocol, our goal is to extract the soft sounds from the cochlea (Figure 3b). We stimulate the cochlea using a train of wideband pulses that are 500 μ s long with a bandwidth of 0-5 kHz. These frequencies are selected as it corresponds to the range important in human hearing. As the responses from the cochlea are extremely weak, we need to combine their responses over time in order to reliably detect their presence.

Please read our paper[1] for other important algorithmic details including a noise detection algorithm for crying sounds, a check to determine if the probe has entered the ear so the system knows when to automatically start the test, and a probe integrity check to detect hardware degradation, which you can find in the paper.

We design a custom low-cost wireless earbud device using off-the-shelf hardware for healthcare workers to perform hearing screening (Figure 4a). Our system has a

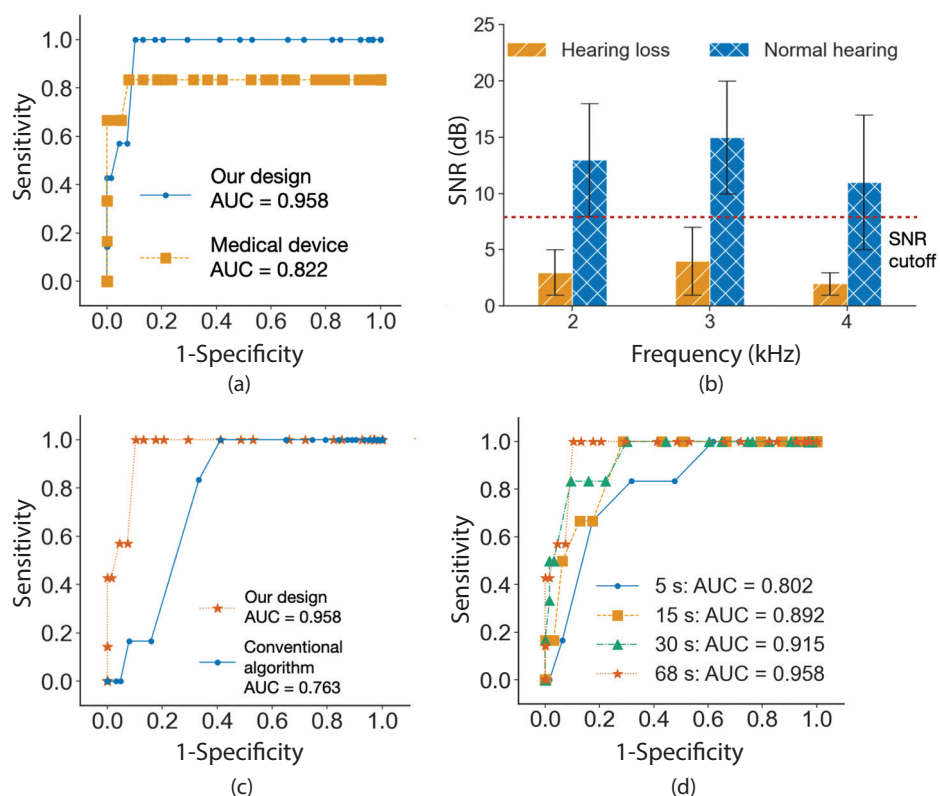


FIGURE 5. System evaluation: (a) Performance of OAEbuds in comparison with commercial OAE medical device. (b) Subgroup analysis comparing the mean SNR of OAEs measured in patients with hearing loss and normal hearing during the clinical study. (c) Comparison of OAEbuds wireless sensing approach with conventional signal processing algorithm on the earbud hardware. (d) Effect of measurement time on clinical performance.

speaker, a microphone, a microcontroller, and a Bluetooth chip, which can stream the acoustic data measured from the ear to a smartphone. On the smartphone we run our algorithms in real-time to detect the weak sounds from the ear. The whole device costs about \$28 each at a quantity of 1000 devices. Our device runs on a 100 mAh Lithium

Polymer Battery. Figure 4b shows that on a single charge, our device can perform up to 90 hearing tests, each lasting for 60 seconds. For shorter test durations of 30 and 15 seconds, our device can perform up to 180 and 360 tests respectively. For context, on a typical day, the hearing loss clinic at our university sees around 20 to 30 patients.

Figure 4c shows that our device can also be recharged via USB when connected to a wall outlet within 3 hours, which can be done overnight in a practical clinic setting.

KEY RESULTS

Here we highlight three key evaluations: First, how well does the system perform for hearing loss screening compared to a commercial medical device. Second, how well does our hearing loss screening algorithms perform compared to prior algorithms on our low-cost hardware? And finally, how does measurement time affect system performance?

We conducted a clinical study across 50 patient ears from two different clinical sites. The attending physician would assign each patient's ground truth hearing status by pooling together results from any previous hearing tests and findings from relevant clinical history. Figure 5a shows the AUC curve of the expensive \$5000 medical device, and the AUC for our system. The figure shows that we obtain comparable performance to an expensive \$5000 medical device but at orders of magnitude lower cost.

To understand this a little more, we perform a subgroup analysis. Figure 5b shows the average SNR levels measured in patients with and without hearing loss. The figure shows that for ears with hearing loss, the average SNR is 3dB across the frequencies. In contrast for ears with normal hearing and functioning cochlea hair cells, the average SNR is 11dB across the frequencies, which exceeds the 8dB SNR threshold that we used for our system. This shows there is a large separation in SNRs between the two classes of ears tested.

In this next result, we demonstrate the importance of hardware-software co-design. To do this, Figure 5c plots the performance of the conventional algorithm used by the expensive medical device when run on our low-cost hardware. This conventional algorithm relies on echo cancellation to remove the reflections from within the ear. The problem however is that this algorithm was designed for on the expensive hardware, and if you instead run it on our low-cost hardware, the signals do not cancel out perfectly, which results in unwanted leftover energy that reduces system performance. And with this setup, we get a low AUC of 0.763. In contrast, when running our

software algorithms on low-cost hardware, we achieve a much higher performance of 0.958. The result demonstrates that the co-design of our low-cost hardware and wireless sensing protocol enables us to obtain high performance while remaining affordable.

Finally, we evaluate the effect of measurement time on system performance. To do this, we set a time limit on the measurement duration and set it to 5 seconds, 15 seconds, 30 seconds, and the maximum measured time of 68 seconds. Figure 5d shows that OAEbuds achieves higher performance than the \$5000 clinical device measurement duration times of 15 and 30 seconds, which is useful for efficient clinical testing.

DISCUSSION AND CONCLUSION

There are several important points to consider when planning future deployments of this system. First, our earbud is designed and tested to work in relatively quiet environments similar to commercial clinical devices. However, for the device to be used in noisier environments, a future design would include passive noise insulating material and an active noise cancellation system to reduce the effect of background noise. Second, regulatory approval would be required to incorporate the system into clinical care. However, we do note that prior FDA clearances of hearing screening devices did not require human testing, which would significantly reduce the cost of regulatory clearance. Third, we note that screening is only the first step in addressing hearing loss. Further followup care and human resources are needed for further diagnostics and treatment. Finally, further field studies should evaluate the use of the device by a range of stakeholders including nurses, technicians, and volunteers in low- and middle-income countries.

To conclude, we present the first pair of wireless earbuds that can perform hearing screening using low-cost hardware. Our clinical study demonstrates similar performance to commercial medical devices and shows the potential of our design to enable hearing loss screening in low- and middle-income countries. If we take a step back beyond this one project, we as a mobile systems community, are uniquely positioned to develop low-cost wearable and mobile technologies[1-4] that can help alleviate global health inequity. ■

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